



Basic Information:

Experiment No: 03

Date:

Title: Sample Java program to check whether the given number is prime palindrome or not.



Objective /Aim:

To develop a Java program that accepts a number from the user and checks whether the given number is Prime, Palindrome, or both Prime and Palindrome, and displays the appropriate result.



System Requirements:

Hardware:

- > Standard PC/Laptop
- > Intel Core i3/i5 processor or equivalent
- > Minimum 4 GB RAM
- > Minimum 500 MB free

Software:

- >Operating System: Windows / Linux / macOS
- > Java Development Kit (JDK 17 or JDK 21)
- > Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- > Command Prompt / Terminal

Algorithm/Logic Description:

- 1.Start
- 2.Read the number n from the user.
- 3.Set count = 0.
- 4.Check all numbers from 1 to n.
 - >If n is exactly divisible by i, increment count.
 - >If count == 2, set prime = true; otherwise set prime = false.
- 5.Store the original number in original.
 - >Set reverse = 0.

->Repeat while $n > 0$:

- Find the last digit using $n \% 10$.
- Add the digit to reverse.
- Remove the last digit using $n / 10$.

6.Compare original with reverse.

->If they are equal, set palindrome = true; otherwise set palindrome = false.

->If both prime and palindrome are true, display Prime Palindrome.

->If only prime is true, display Prime but not Palindrome.

->If only palindrome is true, display Palindrome but not Prime.

7.Otherwise, display Neither Prime nor Palindrome.

8.Close the Scanner.

9.Stop.

UML Class Diagram:

PrimePalindrome

+ main(args : String[]) : void

|Variables used in main(): |

| sc : Scanner |

| n : int |

| count : int |

| prime : boolean |

| original : int |

|reverse : int |

|digit : int |

|palindrome : boolean |

|

| Uses

```

| Scanner |
|         |
| +nextInt() |
| +close() |
|         |

```

Source Code:

```

package record;

import java.util.Scanner;

public class PrimePalindrome {

    public static void main(String[] args) {

        // Create Scanner object to take input from the user

        Scanner sc = new Scanner(System.in);

        // Ask the user to enter a number

        System.out.print("Enter a number: ");

        int n = sc.nextInt();

        // Check whether the number is Prime

        // Variable to count the number of factors

        int count = 0;

        // Check all numbers from 1 to n

        for (int i = 1; i <= n; i++) {

            // If n is exactly divisible by i,

            // then i is a factor of n

            if (n % i == 0) {

                count++;

            }

        }

        // Variable to store whether the number is prime

        boolean prime;

```

```

// A prime number has exactly two factors:
// 1 and the number itself
    if (count == 2) {
        prime = true;
    } else {
        prime = false;
    }

// Check whether the number is Palindrome
// Store the original number
// because n will be changed while reversing
    int original = n;

// Variable to store the reversed number
    int reverse = 0;

// Reverse the number using while loop
    while (n > 0) {
        // Get the last digit of the number
        int digit = n % 10;

        // Add the digit to the reversed number
        reverse = reverse * 10 + digit;

        // Remove the last digit from n
        n = n / 10;
    }

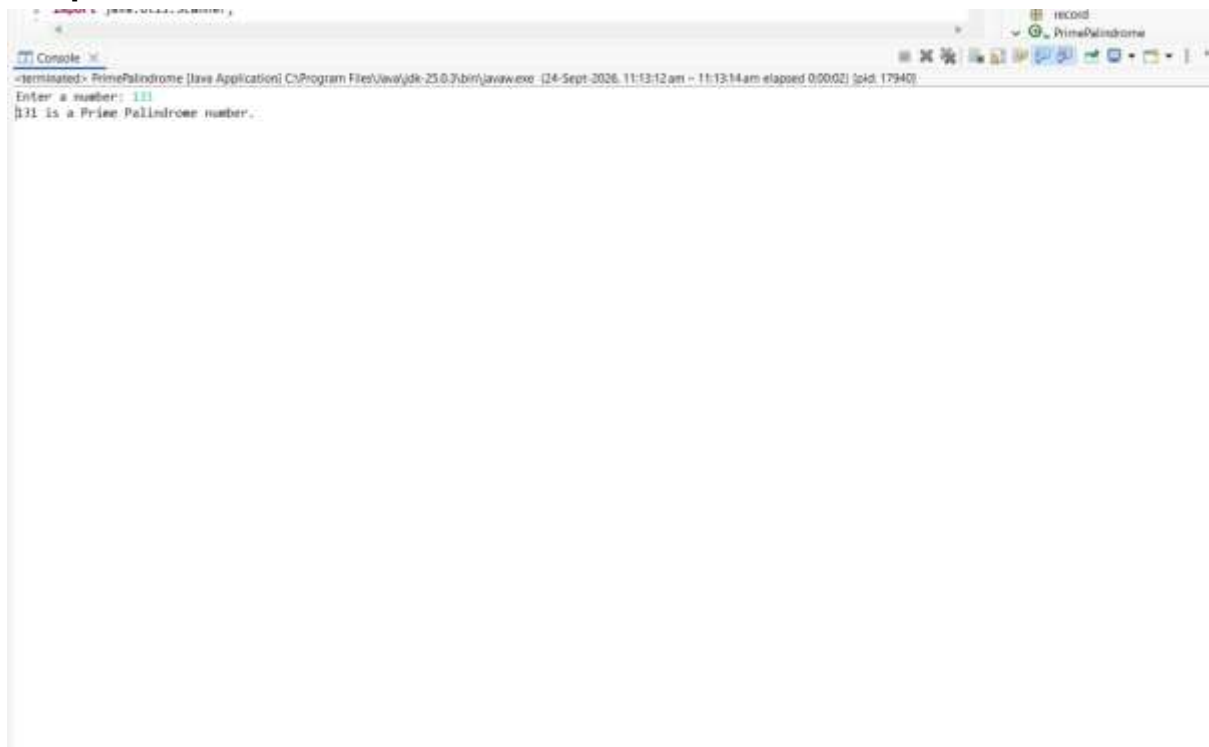
// Variable to store whether the number is palindrome
    boolean palindrome;

// Compare original number with reversed number
    if (original == reverse) {
        palindrome = true;
    } else {
        palindrome = false;
    }

```

```
}  
  
    // Display the Result  
  
    // Check if the number is both Prime and Palindrome  
  
        if (prime && palindrome) {  
  
            System.out.println(original +" is a Prime Palindrome number.");  
  
        // Check if the number is Prime but not Palindrome  
  
        } else if (prime) {  
  
            System.out.println(original +" is Prime but not Palindrome.");  
  
        // Check if the number is Palindrome but not Prime  
  
        } else if (palindrome) {  
  
            System.out.println(original +" is Palindrome but not Prime.");  
  
        // If the number is neither Prime nor Palindrome  
  
        } else {  
  
            System.out.println(original + " is neither Prime nor Palindrome.");  
  
        }  
  
    // Close the Scanner object  
  
    sc.close();  
  
}  
  
}
```

Output Screenshot:



Result:

The PrimePalindrome Java program was successfully implemented and executed. The program accepts an integer from the user and performs two checks: whether the given number is a prime number and whether it is a palindrome number. The program first counts the factors of the number to determine whether it is prime. It then reverses the number and compares the reversed value with the original value to determine whether it is a palindrome. Finally, based on both conditions, the program displays whether the number is a Prime Palindrome, Prime but not Palindrome, Palindrome but not Prime, or neither Prime nor Palindrome.